■ Developer Mock Test Results

Test ID:	13ed9492-38e2-41...	**Student ID:**	52086
Test Type:	Developer Assessment	**Date:**	2026-03-04 15:37:53
Overall Score:	0/25 (0.0%)	**Performance:**	■■ Needs Improvement
Warnings:	0/3	**Status:**	Completed ■

■ Section-wise Performance

Section	Score	Percentage	Status
■ Aptitude	0/10	0.0%	■■ Needs Work
■ MCQ/Theory	0/10	0.0%	■■ Needs Work
■ Coding	0/5	0.0%	■■ Needs Work

■ Detailed Question Review

■ APTITUDE SECTION (0/10 - 0.0%)

■ Q1. If 2 men can build a wall in 12 days, how many men are needed to build the same wall in 6 days?

Your Answer: No answer (Skipped)
Correct Answer: 4
■ *To find the number of men needed to build the wall in 6 days, we can use the formula: men * days = constant. Given that 2 men can build the wall in 12 days, we have 2 * 12 = 24. To build the same wall in 6 days, we set up the equation: men * 6 = 24. Solving for men, we get men = 24 / 6 = 4. The correct answer is indeed 4, as calculated. If the user got it wrong, their likely mistake was not using the inverse relationship between the number of men and the number of days to solve the problem.*

■ Q2. The average of 8 numbers is 12. If one of the numbers is 15, what is the sum of the remaining 7 numbers?

Your Answer: No answer (Skipped)
Correct Answer: 80
■ *To find the sum of the remaining 7 numbers, we first calculate the total sum of all 8 numbers, which is the average (12) multiplied by the number of numbers (8), giving us 12 * 8 = 96. If one of the numbers is 15, we subtract this from the total sum to find the sum of the remaining 7 numbers: 96 - 15 = 81. However, the correct answer provided is 80, suggesting a possible miscalculation or misunderstanding in the given solution. Given the calculation, the user's answer or the provided correct answer might have a discrepancy. The calculation clearly shows that the sum of the remaining 7 numbers should be 81, not 80. Therefore, the likely mistake is in the provided correct answer or in interpreting the question's data.*

■ Q3. A product is on sale for 20% off its original price. If the original price is $100, how much will you pay for the product after the discount?

Your Answer: No answer (Skipped)
Correct Answer: $80

■ *To find the discounted price, calculate 20% of the original price ($100) and subtract it from the original price. The key calculation step is: $100 * 0.20 = $20, then $100 - $20 = $80. Therefore, the correct answer is $80, which matches the given correct answer, confirming that the user's answer is correct.*

■ Q4. If a number is decreased by 20% and then increased by 20%, what is the overall percentage change?

Your Answer: No answer (Skipped)

Correct Answer: -4%

■ *To find the overall percentage change, let's consider a number, say 100. Decreasing it by 20% gives 80, and then increasing 80 by 20% gives 96. The key calculation step is: 80 * 1.20 = 96. Since 96 is 4 less than the original 100, the overall percentage change is -4%. The correct answer is indeed D) -4%. If the user got it wrong, their likely mistake was not considering the effect of the percentage changes sequentially, or not calculating the final result correctly.*

■ Q5. If 8 workers can build a house in 12 days, how many days will it take for 6 workers to build the same house?

Your Answer: No answer (Skipped)

Correct Answer: 16 days

■ *To find the number of days it will take for 6 workers to build the house, we can use the concept of work and time inversely related to the number of workers. The key calculation step is: 8 workers * 12 days = 6 workers * x days, which gives 96 = 6x. Solving for x, we get x = 96 / 6 = 16 days. This calculation confirms that the correct answer is indeed 16 days, as the amount of work required to build the house remains constant, and the product of the number of workers and the days to complete the work should also remain constant.*

■ Q6. A mixture is made of 2 parts acid and 3 parts water. What is the ratio of acid to the total mixture?

Your Answer: No answer (Skipped)

Correct Answer: 2/5

■ *The correct answer is 2/5 because the mixture is made of 2 parts acid and 3 parts water, making a total of 2 + 3 = 5 parts. The ratio of acid to the total mixture is then 2 (acid parts) / 5 (total parts) = 2/5. The user likely got it wrong by misinterpreting the ratio or forgetting to calculate the total parts of the mixture.*

■ Q7. A mixture of paint and water is in the ratio 3:4. If 3 cups of paint are added to the mixture, what is the new ratio of paint to water?

Your Answer: No answer (Skipped)

Correct Answer: 6:4

■ *To find the new ratio, let's assume the initial mixture has 3x cups of paint and 4x cups of water. When 3 cups of paint are added, the total paint becomes 3x + 3. Since the initial ratio of paint to water is 3:4, we can set up a proportion, but a key step is to find a common multiplier for the initial ratio, which is not directly provided. However, given the answer choices, let's analyze the correct answer: if the new ratio is 6:4, this implies the total paint is now 6 parts and the water remains 4 parts (since no water was added). The likely mistake for an incorrect answer would be not properly accounting for the added paint in relation to the initial ratio. The correct calculation step involves setting up an equation based on the ratio, but without a specific initial amount, we consider the ratio directly. The correct answer, 6:4, suggests that*

■ Q8. What is the next number: 1, 3, 6, 10, 15, ?

Your Answer: No answer (Skipped)

Correct Answer: 21

■ *The correct answer is 21. This sequence appears to be formed by adding 2, 3, 4, 5, and so on to the previous term, which is a characteristic of triangular numbers. The key calculation step is: 1 + 2 = 3, 3 + 3 = 6, 6 + 4 = 10, 10 + 5 = 15, and 15 + 6 = 21, confirming that the next number in the sequence is indeed 21.*

■ Q9. A shirt is on sale for 15% off its original price of $80. How much will you pay for the shirt?

Your Answer: No answer (Skipped)

Correct Answer: $68

■ *To find the sale price, calculate the discount amount by multiplying the original price by the discount percentage: $80 * 0.15 = $12. Then, subtract the discount from the original price: $80 - $12 = $68. The correct answer is $68, which matches the calculation, so the user's answer is correct.*

■ Q10. If it is true that all A are B and some B are not C, what can be concluded about A and C?

Your Answer: No answer (Skipped)
Correct Answer: Some A are C
■ *The correct answer is "Some A are C" because if all A are B and some B are not C, it implies that not all B are C, but it doesn't rule out the possibility that some B (and therefore some A) are C. The user's answer is not provided, but a likely mistake would be assuming that since some B are not C, then no A can be C, which is a flawed conclusion. The given information only guarantees that A is a subset of B, but it doesn't provide enough information to determine the exact relationship between A and C.*

■ MCQ SECTION (0/10 - 0.0%)

■ Q1. What is profiling in Java?

Your Answer: No answer (Skipped)
Correct Answer: The process of analyzing the performance of a program
■ *The correct answer is A) The process of analyzing the performance of a program. This is because profiling in Java refers to the process of analyzing and measuring the performance of a Java program, identifying areas where the program is spending most of its time, and optimizing those areas to improve overall performance. The user's likely mistake is confusing profiling with other types of analysis, such as functionality or security testing, which are distinct activities with different goals.*

■ Q2. What is starvation in Java?

Your Answer: No answer (Skipped)
Correct Answer: A situation where a thread is unable to access a shared resource
■ *The correct answer is A) A situation where a thread is unable to access a shared resource. This is because starvation in Java refers to a scenario where a thread is unable to gain access to a shared resource due to other threads holding onto it for an extended period, preventing the starved thread from making progress. The user's answer is correct, so there is no likely mistake to explain.*

■ Q3. What type of programming language is Java?

Your Answer: No answer (Skipped)
Correct Answer: High-level, object-oriented
■ *The correct answer is B) High-level, object-oriented because Java is a language that abstracts away many low-level details, allowing developers to focus on the logic of their program without worrying about memory management, making it a high-level language. Additionally, Java is based on the principles of object-oriented programming, which organizes code into objects that contain data and functions that operate on that data. The user's likely mistake would be confusing Java with a low-level language like C or C++, which require manual memory management and are not exclusively object-oriented.*

■ Q4. What is the file extension used for Java programs?

Your Answer: No answer (Skipped)
Correct Answer: .java
■ *The correct answer is A) .java because Java programs are typically written and saved with this file extension, which allows them to be compiled into bytecode. The .java file contains the source code of the program, making it the primary file extension associated with Java programming. The user likely got it wrong if they confused .java with .class, which is the file extension used for compiled Java bytecode, or .jar, which is used for packaged Java archives.*

■ Q5. What is the purpose of data types in Java?

Your Answer: No answer (Skipped)
Correct Answer: To define the type of data
■ *The correct answer is D) To define the type of data. This is because data types in Java determine the type of value a variable can hold, such as integer, character, or string, which helps ensure that the correct operations are performed on the data. The user's answer is correct, so there is no likely mistake to explain, but if they had chosen A) To store data values, they might have been confusing the purpose of variables with the purpose of data types.*

■ Q6. What type of programming concepts does Java mainly follow?

Your Answer: No answer (Skipped)
Correct Answer: Object-oriented programming

■ *The correct answer is B) Object-oriented programming because Java is designed to support the principles of object-oriented programming (OOP), such as encapsulation, inheritance, and polymorphism. Java's syntax and features, like classes, objects, and methods, are all centered around OOP concepts. The user's likely mistake, if they chose a different option, would be misunderstanding Java's fundamental programming paradigm, which is rooted in object-oriented design.*

■ Q7. What is a best practice for writing Java code?

Your Answer: No answer (Skipped)
Correct Answer: Follow Java coding standards
■ *The correct answer is A) Follow Java coding standards. This is because following Java coding standards ensures that the code is readable, maintainable, and efficient, making it easier for other developers to understand and work with. By following these standards, developers can avoid common mistakes and write high-quality code, which is essential for building reliable and scalable software systems. (Note: Since this is not a math question, there are no calculations to show.)*

■ Q8. What is a deadlock in Java?

Your Answer: No answer (Skipped)
Correct Answer: A situation where two or more threads are blocked indefinitely
■ *The correct answer is A) A situation where two or more threads are blocked indefinitely. This is because a deadlock in Java occurs when two or more threads are unable to proceed with their execution due to a circular dependency on resources, resulting in a permanent blockage. The user likely chose the correct answer, but if they didn't, their mistake was probably due to misunderstanding the concept of deadlock, which involves multiple threads and a circular wait, rather than a single thread or concurrent execution.*

■ Q9. What is debugging in Java?

Your Answer: No answer (Skipped)
Correct Answer: The process of finding and fixing errors in code
■ *The correct answer is A) The process of finding and fixing errors in code. This is because debugging specifically refers to the process of identifying and correcting errors, or bugs, in the code that prevent it from functioning as intended. The user's answer is correct, so there is no likely mistake to explain, but it's worth noting that options B, C, and D are close, but not entirely accurate, as debugging is primarily focused on fixing errors, not just exceptions or all types of problems.*

■ Q10. What is the design of Java programs that makes them easy to maintain?

Your Answer: No answer (Skipped)
Correct Answer: Structured and class-based
■ *The correct answer is B) Structured and class-based because Java programs are designed using a structured approach, which organizes code into logical modules, and a class-based approach, which allows for the creation of reusable objects. This design makes Java programs easy to maintain by providing a clear and modular structure. The user's likely mistake is not recognizing the importance of structure and classes in Java programming, which are fundamental concepts that enable maintainable code.*

■ CODING SECTION (0/5 - 0.0%)

■ Q1. Write a program that reads a list of integers and prints the filtered list of even numbers. **Input Format:** - Line 1: Number of integers (integer) ...

Your Answer:

```
No answer (Skipped)
```

Correct Answer:

```
n = int(input())
numbers = list(map(int, input().split()))
even_numbers = [num for num in numbers if num % 2 == 0]
print(*even_numbers)
```

■ *There are no test case results provided to analyze the issue with the student's code. No feedback can be given without the actual test case results. Please provide the test case results to identify the problem.*

■ **Q2. Write a program that reads a string and prints the string with all characters converted to uppercase. **Input Format:** - Line 1: Input string (strin...**

Your Answer:

```
No answer (Skipped)
```

Correct Answer:

```
print(input().upper())
```

■ *There is no test case failure to analyze, as no test case results or student code were provided. No feedback can be given without this information. Please provide the test case results to determine the issue.*

■ **Q3. Write a Java program that reads a string from stdin and decrypts it using a simple substitution cipher. **Input Format:** - Line 1: A string **Outpu...**

Your Answer:

```
No answer (Skipped)
```

Correct Answer:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        String input = scanner.nextLine();
        String decrypted = decrypt(input);
        System.out.println(decrypted);
        scanner.close();
    }

    public static String decrypt(String input) {
        StringBuilder decrypted = new StringBuilder();
        for (char c : input.toCharArray()) {
            if (c == 'a') {
                decrypted.append('b');
            } else if (c == 'b') {
                decrypted.append('a');
            } else if (c == 'c') {
                decrypted.append('d');
            } else if (c == 'd') {
                decrypted.append('c');
            } else if (c == 'e') {
                decrypted.append('f');
            } else if (c == 'f') {
                decrypted.append('e');
            } else if (c == 'g') {
                decrypted.append('h');
            } else if (c == 'h') {
                decrypted.append('g');
            } else if (c == 'i') {
                decrypted.append('j');
            } else if (c == 'j') {
                decrypted.append('i');
            } else if (c == 'k') {
                decrypted.append('l');
            } else if (c == 'l') {
                decrypted.append('k');
            } else if (c == 'm') {
                decrypted.append('n');
            } else if (c == 'n') {
                decrypted.append('m');
            } else if (c == 'o') {
                decrypted.append('p');
```

```
            } else if (c == 'p') {
                decrypted.append('o');
            } else if (c == 'q') {
                decrypted.append('r');
            } else if (c == 'r') {
                decrypted.append('q');
            } else if (c == 's') {
                decrypted.append('t');
            } else if (c == 't') {
                decrypted.append('s');
            } else if (c == 'u') {
                decrypted.append('v');
            } else if (c == 'v') {
                decrypted.append('u');
            } else if (c == 'w') {
                decrypted.append('x');
            } else if (c == 'x') {
                decrypted.append('w');
            } else if (c == 'y') {
                decrypted.append('z');
            } else if (c == 'z') {
                decrypted.append('y');
            } else {
                decrypted.append(c);
            }
        }
        return decrypted.toString();
    }
}
```

■ *Since no student code was submitted, there's no code to analyze. The test case results show an expected output of "hello" for the input "khoor", but without the student's code, it's impossible to determine why the output might differ from the expected result. The student needs to submit their code to receive accurate feedback.*

■ Q4. Write a program that reads two integers and prints their sum, difference, product, and quotient. **Input Format:** - Line 1: First integer (integer) ...

Your Answer:

```
No answer (Skipped)
```

Correct Answer:

```
a = int(input())
b = int(input())

print(a + b, a - b, a * b, a / b)
```

■ *There is no test case failure information provided to analyze the issue. No code was submitted, and no test case results are available to determine what went wrong. Without this information, it's impossible to provide accurate feedback.*

■ Q5. Write a Java program that defines a Shape interface with methods to calculate area and perimeter, and a Circle class that implements Shape. **Input F...

Your Answer:

```
No answer (Skipped)
```

Correct Answer:

```
import java.util.Scanner;

interface Shape {
    double calculateArea();
    double calculatePerimeter();
}
```

```
class Circle implements Shape {
    private double radius;

    public Circle(double radius) {
        this.radius = radius;
    }

    @Override
    public double calculateArea() {
        return Math.PI * Math.pow(radius, 2);
    }

    @Override
    public double calculatePerimeter() {
        return 2 * Math.PI * radius;
    }
}

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int n = scanner.nextInt();
        scanner.nextLine(); // Consume newline left-over

        for (int i = 0; i < n; i++) {
            String shapeType = scanner.next();
            double radius = scanner.nextDouble();
            scanner.nextLine(); // Consume newline left-over

            if (shapeType.equals("Circle")) {
                Circle circle = new Circle(radius);
                System.out.printf("%.1f %.1f%n", circle.calculateArea(),
circle.calculatePerimeter());
            } else if (shapeType.equals("Shape")) {
                // Assuming Shape with radius is a circle
                Circle circle = new Circle(radius);
                System.out.printf("%.1f %.1f%n", Math.PI * Math.pow(radius, 2), 2 * Math.PI *
radius);
            }
        }

        scanner.close();
    }
}
```

■ *There is no test case failure provided to analyze. The student's code and test case results are not shown, making it impossible to determine what went wrong. Please provide the test case results to give accurate feedback.*

■ Recommendations

Areas that need improvement:

- **Aptitude:** Practice more logical reasoning, quantitative aptitude, and problem-solving questions.
- **MCQ/Theory:** Review programming concepts, data structures, and algorithms theory.
- **Coding:** Practice more coding problems on platforms like LeetCode or HackerRank.
